Folkeskole Cascade — Edge Table (DRAFT for review)

Purpose. This is the cited evidence base for the education cascade card in the WELA apparatus. It mirrors WELA's own flagship argument: underfunding the folkeskole produces consequences that GDP is blind to but that cascade through learning, labour-market attachment, income, health, wellbeing, and inequality.

This is Track 1 — the gate. Nothing lands in `data/relationships.json` until you've read this and resolved the `NEEDS` rows. Treat every ⚠ `NEEDS` as a side-task for you: either confirm the number from memory/expertise, or hand me the paper (ideally into a NotebookLM notebook so I can extract the grounded figure without hallucination risk).

How to read the table.

- `direction`: `positive` = the two move the same way after sign-harmonisation per the `relationships.json` convention (see its `_examples`); `negative` = they move oppositely. I've added a plain-language **"effect"** column so the sign convention never bites us — trust the plain-language one, I'll encode the JSON sign correctly from it.
- `strength`: qualitative 0–1 ordering only (NOT a calibrated coefficient). 0.2 weak, 0.5 moderate, 0.8 strong. This is the field most likely to draw a "where's that from?" from Ploug/Falkenfleth — so it's deliberately labelled qualitative everywhere.
- `lag_years`: rough years to materially register.
- `confidence`: my confidence in the *existence and direction* of the edge, given established literature — `high` = textbook causal consensus, `med` = well-evidenced but contested magnitude, `low` = plausible/mechanism-clear but thin direct evidence.
- ⚠ `NEEDS`: I'm leaning on recall for the *number* and want it hardened before Friday.

The lever

`folkeskole_funding` is **not yet a node** in `relationships.json` (the 26 nodes are all measured indicators). For the cascade we introduce it as a **policy-lever node** — the thing WELA's report is actually about — that feeds the measured nodes. Flagging this as decision **D1** below.

Core cascade edges (the spine — folkeskole → outcomes)

#	from	to	effect (plain language)	dir	strength	lag (yrs)	conf	source	flag
1	folkeskole_funding	learning_outcomes	More per-pupil funding → higher learning outcomes (test scores)	+	0.4	3	med	Hanushek & Woessmann, *The Knowledge Capital of Nations* (2015); meta-evidence on funding→achievement is mixed on magnitude but positive when targeted	⚠ NEEDS exact effect size; funding→achievement magnitude is genuinely contested
2	folkeskole_funding	neet	Underfunding → weaker support/retention → more youth not in education/employment/training	−	0.35	5	med	OECD *Education at a Glance 2023*, NEET chapter; mechanism via early-leaving	⚠ NEEDS DK-specific link if available
3	learning_outcomes	income	Higher cognitive skills/attainment → higher lifetime earnings (human capital)	+	0.6	8	high	Mincer earnings function; OECD EaG returns-to-education; ~8–10% wage premium per school year	⚠ NEEDS confirm the per-year premium figure (Mincer / OECD)
4	learning_outcomes	unemployment	Higher skills → lower unemployment risk	−	0.4	5	high	OECD EaG (employment rates by attainment); ILO	—
5	neet	income	Youth detachment → scarring → persistently lower lifetime income	−	0.5	10	high	Gregg & Tominey (2005) youth unemployment wage-scarring; Eurofound NEET cost studies	⚠ NEEDS the scarring magnitude (Gregg/Tominey or Eurofound)
6	neet	life_satisfaction	Detachment → lower subjective wellbeing	−	0.35	2	med	Clark et al., *The Origins of Happiness* (2018); unemployment→life-satisfaction is one of the most robust	⚠ NEEDS effect size

#	from	to	effect (plain language)	dir	strength	lag (yrs)	conf	source	flag
								findings in wellbeing economics	

Second-order edges (income/skills → broader wellbeing & equity)

#	from	to	effect (plain language)	dir	strength	lag (yrs)	conf	source	flag
7	income	life_satisfaction	Higher income → higher life satisfaction (diminishing returns)	+	0.3	1	high	Kahneman & Deaton (2010); Killingsworth (2021); concave, not linear	—
8	income	hale	Higher income → better health / healthy life expectancy (social gradient)	+	0.4	10	high	Marmot, *The Health Gap* (2015); WHO social determinants	⚠ NEEDS DK/Nordic gradient figure if you want it country-specific
9	income	life_expectancy	Same social gradient, on total life expectancy	+	0.4	10	high	Marmot; Chetty et al. (2016) income–longevity gradient	—
10	learning_outcomes	gini	Broad-based education attainment → compresses the wage distribution → lower inequality	−	0.3	15	med	Goldin & Katz, *The Race between Education and Technology* (2008)	⚠ NEEDS — this is the most contested edge (education↔inequality direction debated); flag low-ish
11	neet	poverty_rate	Youth detachment → higher poverty risk	+	0.4	5	med	Eurofound *NEETs* (2012, 2016)	⚠ NEEDS DK figure
12	gini	life_satisfaction	Higher inequality → lower average life satisfaction (society-level)	−	0.25	3	low	Wilkinson & Pickett, *The Spirit Level* (2009) — influential but contested; keep confidence low	⚠ NEEDS — contested; decide whether to include (D2)

Reinforcing / loop edges (optional — shows it's a system, not a chain)

#	from	to	effect (plain language)	dir	strength	lag (yrs)	conf	source	flag
13	income	learning_outcomes	Higher household income → better child learning outcomes (next generation) → feedback loop	+	0.35	15	med	Intergenerational mobility; Chetty et al. (2014); the loop that makes underfunding compounding	⚠ NEEDS — include only if we want the reinforcing loop (D3)

Decisions I need from you (blocking the JSON populate)

- **D1 — the lever node.** Add `folkeskole_funding` as an explicit policy-lever node (distinct styling from measured indicators), OR start the cascade at `learning_outcomes` and treat "underfunding" as narrative text only? **My rec: add the lever node** — it's WELA's actual subject and makes the "what would we do" legible. Without it the graph starts mid-story.
- **D2 — edge 12 (inequality→life satisfaction).** Contested (*Spirit Level*). Include at low confidence with a visible caveat, or cut? **My rec: include, low confidence, dashed** — it's exactly the kind of edge WELA will want to argue about, which is the point.
- **D3 — edge 13 (the reinforcing loop).** Include the intergenerational feedback loop, or keep the graph a clean DAG for v1? **My rec: include** — "underfunding compounds across generations" is a strong WELA-aligned message and visually shows this is a *system*.
- **D4 — country-specificity.** Several edges have Danish/Nordic-specific figures available (social gradient, NEET cost). Do you want me to push for DK-specific numbers (more credible to this audience, more research load), or ship with general OECD/literature figures labelled as such for v1? **My rec: general for v1, DK-specific as the post-meeting hardening pass** — the *gaps panel* already says "we'd want DK-calibrated edges," which turns this limitation into part of the testing story.

The 8 ⚠ NEEDS rows — your side-task summary

If you can hand me papers (or drop them in NotebookLM) for these, the card goes from "plausible" to "cited" for Friday. Priority order:

1. **#3** education→income per-year wage premium (Mincer/OECD) — *highest value, most likely to be challenged*
2. **#5** NEET→income scarring magnitude (Gregg & Tominey / Eurofound)
3. **#1** funding→learning-outcomes effect size (Hanushek/Woessmann) — *contested, worth nailing*
4. **#10** education→inequality (Goldin & Katz) — *most contested direction*

5. **#6** unemployment/NEET→life-satisfaction effect (Clark et al.)
6. **#8** income→health gradient, DK if possible (Marmot/Chetty)
7. **#2, #11** DK-specific NEET links (OECD EaG / Eurofound)

Everything else (#4, #7, #9) I'm confident enough to ship without a hardening pass.